

CubeSat Low-Power Control Unit

Technical Design Document

Challenge:	Challenge 1 — Sustainable Electronics for Space Systems
Event:	AESS Sustainability Hackathon 2026
Subsystem:	Attitude & Health Monitor Control Unit (LPCU)
Platform:	ESP32 + MPU-6050 + TMP117 3.3 V Deep Sleep
Mission:	1U CubeSat in Low Earth Orbit (LEO), 90-min orbit
Strategy:	Duty cycling + deep sleep 3.3% active duty cycle

96.6% Power reduction	277.1 mW → 9.5 mW Average system power	×29.3 lifetime 0.06 → 1.76 days
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1. Mission scenario and subsystem definition

The LPCU is designed for a 1U CubeSat operating in a low Earth orbit at approximately 500 km altitude with an orbital period of 90 minutes. About 35% of each orbit is spent in eclipse, where no solar power is available. The satellite carries a passive magnetic attitude-stabilisation system and requires periodic health telemetry throughout its mission.

The LPCU performs three functions: reading 6-axis inertial data from the MPU-6050 IMU, reading board temperature from the TMP117 sensor, and assembling and emitting a compact telemetry frame over UART. One measurement cycle per minute is sufficient for passive-stabilisation monitoring — this observation is the foundation of the duty-cycle strategy.

Why this subsystem?

Attitude and health monitoring is the lowest-frequency, highest-value housekeeping task on a small satellite. A conventional always-on design runs the MCU at full speed continuously, even though useful work occurs for less than 4% of the operating period. That gap is the optimisation target.

2. System architecture

Functional blocks:

Block	Component	Role
Power source	Solar panel + 0.4 Wh Li-ion	Primary and backup power
Voltage reg.	TPS63021 (buck-boost)	3.3 V regulated rail, 60 μ A I _q
MCU	ESP32 (240 MHz, dual-core)	Processing, I2C master, UART, timer
IMU	MPU-6050 (accel + gyro)	6-DOF attitude data via I2C
Temp sensor	TMP117 (one-shot mode)	Board temperature, $\pm 0.1^{\circ}\text{C}$ accuracy
Power control	ESP32 internal ULP timer	58 s deep-sleep wakeup cycle
Telemetry out	UART frame → RF transceiver	Attitude + health packet per cycle

The I2C bus runs at 400 kHz. The TPS63021 regulator was selected for its 60 μ A quiescent current and wide input range (1.8–5.5 V), compatible with both fully-charged and partially-discharged Li-ion cells. The ESP32 ULP (ultra-low-power) co-processor maintains the 58-second timer during deep sleep without waking the main cores.

3. Power budget — baseline vs optimized

Baseline design (always-on, no sleep management):

Component	Mode	Current (mA)	Voltage (V)	Power (mW)
ESP32 MCU	Active @ 240 MHz	80.00	3.3	264.00
MPU-6050 IMU	Measurement	3.40	3.3	11.22

TMP117 temp	Continuous	0.50	3.3	1.65
TPS63021 vreg	Always on	0.06	3.3	0.20
TOTAL	—	83.96	3.3	277.07

Optimized design (duty cycle 3.3%, deep sleep):

Duty cycle: 2 s active measurement window / 60 s cycle period = 3.33%

Component	Active (mW)	Sleep (mW)	Avg (mW)	Method
ESP32 MCU	264.00	0.033	8.79	Deep sleep, ULP timer
MPU-6050 IMU	11.22	0.017	0.39	Cycle mode standby
TMP117 temp	1.65	0.005	0.06	One-shot, then shutdown
TPS63021 vreg	0.20	0.20	0.20	Always on (Iq only)
TOTAL	277.07	0.255	9.44	96.6% below baseline

Energy and lifetime projection:

Metric	Baseline	Optimized	Improvement
Avg system power	277.1 mW	9.5 mW	96.6% reduction
Energy per orbit	0.416 Wh	0.014 Wh	96.6% less
Energy per day	6.65 Wh	0.23 Wh	96.6% less
Days on 0.4 Wh budget	0.06 days	1.76 days	×29.3 longer
Orbits on 0.4 Wh	1.0 orbits	28.2 orbits	×29.3 longer

4. Power optimisation strategy

Strategy: deep sleep + duty cycling

The ESP32 enters deep sleep (10 μ A) immediately after each telemetry frame is transmitted. The ULP co-processor keeps a 58-second hardware timer running at negligible cost. On wakeup, the main cores boot, perform one measurement cycle (2 seconds), transmit the frame, and return to sleep. This produces a 3.3% duty cycle.

Each peripheral is placed in its lowest-power state during the sleep window: the MPU-6050 enters cycle mode (5 μ A), and the TMP117 is commanded to one-shot mode so it only draws power during the 15.5 ms conversion period. The voltage regulator remains active but its quiescent current (60 μ A) is already the minimum possible for this rail architecture.

Average power derivation:

$$P_{\text{avg}} = P_{\text{active}} \times D + P_{\text{sleep}} \times (1 - D)$$

For the ESP32: $P_{\text{avg}} = 264.0 \text{ mW} \times 0.033 + 0.033 \text{ mW} \times 0.967 = 8.76 \text{ mW} + 0.032 \text{ mW} = 8.79 \text{ mW}$

The MCU dominates total power in both modes, which is why ESP32 deep sleep is the single most impactful technique — it removes 255.2 mW from the 97% of time when no processing is needed.

5. Component selection justification

Component	Key spec	Why selected
ESP32	Deep sleep: 10 μ A; ULP co-proc	Only MCU class with hardware sleep this deep AND a secondary processor that maintains timers without waking main cores. Widely used in space-adjacent applications.
MPU-6050	Cycle mode: 5 μ A; full: 3.4 mA	Mature, well-characterised IMU with hardware standby mode. Datasheet-confirmed sleep current. Attitude-grade performance for passive stabilisation.
TMP117	One-shot: 1.5 μ A standby; $\pm 0.1^{\circ}\text{C}$	Purpose-built for duty-cycled use. One-shot mode draws power only during conversion (15.5 ms). Best-in-class accuracy for thermal monitoring.
TPS63021	Iq: 60 μ A; range: 1.8–5.5 V in	Buck-boost topology handles battery voltage variations across charge cycle. 60 μ A quiescent is among the lowest available in this form factor.

6. Sustainability impact

In a LEO CubeSat, the subsystem power budget is a hard constraint. The total satellite power budget for a 1U satellite typically ranges from 1 W to 5 W across all subsystems. A housekeeping unit consuming 277 mW continuously would consume 5–28% of the entire satellite budget for a task that requires active processing only 3.3% of the time.

The optimised LPCU at 9.5 mW releases approximately 267 mW back to the mission — enough to power a UHF radio beacon, an additional science sensor, or extend the satellite's eclipse-survival window during anomalies.

Longer mission lifetime directly reduces the environmental cost per measurement: the same launch mass and manufacturing energy yields 29 $\times$ more data. From a lifecycle sustainability perspective, extending useful satellite life is one of the most significant levers available to space system designers.

Impact dimension	Baseline	Optimized	Benefit
Power budget consumed	277.1 mW	9.5 mW	267 mW freed for mission
Thermal dissipation	≈ 0.28 W	≈ 0.01 W	Lower thermal stress
Battery runtime	0.06 days	1.76 days	Eclipse anomaly survival
Data per launch energy	1 $\times$	29 $\times$	Better lifecycle value

All power values derived from component datasheets and verified with the Python simulation in /simulation/power_model.py. AI tools (Claude) were used for documentation drafting. All calculations and engineering decisions were validated by the team.